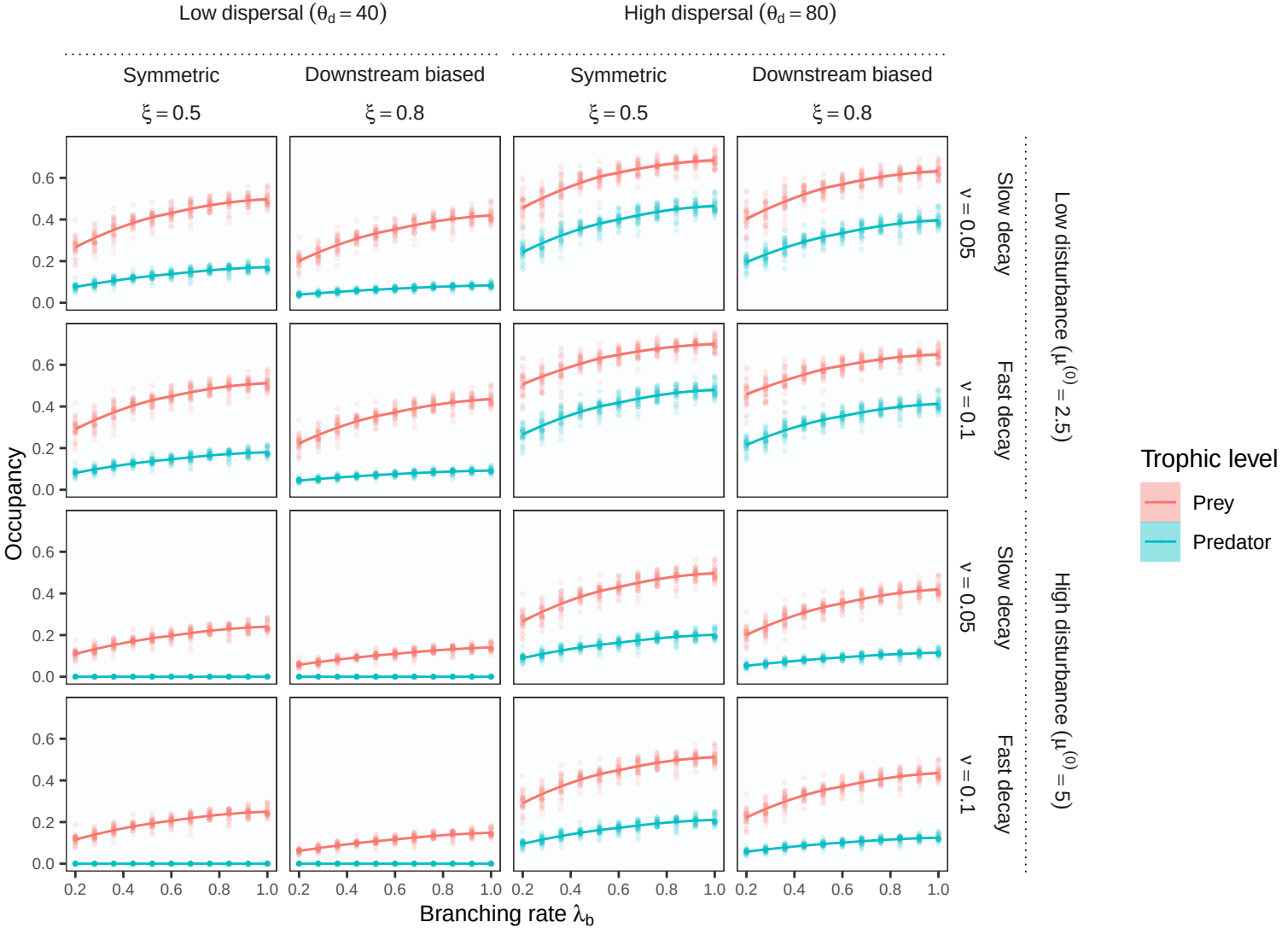

Low dispersal ($\theta_d = 40$)
High dispersal ($\theta_d = 80$)
Symmetric
Downstream biased
Symmetric
Downstream biased
$\xi = 0.5$
$\xi = 0.8$
$\xi = 0.5$
$\xi = 0.8$
Slow decay
Fast decay
Slow decay
Fast decay
$\nu = 0.05$
$\nu = 0.1$
$\nu = 0.05$
$\nu = 0.1$
Low disturbance ($\mu^{(0)} = 2.5$)
High disturbance ($\mu^{(0)} = 5$)
Occupancy
0.0
0.2
0.4
0.6
0.2
0.4
0.6
0.8
1.0
Branching rate λ_b
Trophic level
Prey
Predator